Institutional Repositories: Importance, Implementation, and Impact

Dr. M. Krishnamurthy
Professor and Head
Documentation Research and Training Centre,
Indian Statistical Institute,
Chair- Center for Academica and Industry Research
- ResearcherID: O-3809-2018
oorcid.org/0000-0002-8578-1533
mkrishnamurthy1304@gmail.com

- DSpace is an open-source software platform that enables organisations to:
- capture and describe digital material using a submission workflow module, or a variety of programmatic ingest options
- distribute an organisation's digital assets over the web through a search and retrieval system
- preserve digital assets over the long term

Introduction

What is IR?

- An **institutional repository** (**IR**) is an archive for collecting, preserving, and disseminating digital copies of the intellectual output of an institution, particularly a research institution.
- Academics also utilize their IRs for archiving published works to increase their visibility and collaboration with other academics.

What are Repositories?

- Platforms for storing, preserving, and sharing open data.

Relevance to Libraries

- Libraries as facilitators of knowledge access and custodians of digital data.

Cont…

- In 2003, the functions of an institutional repository were described by **Clifford Lynch** in relation to universities. He stated that:

"... a university-based institutional repository is a set of services that a university offers to the members of its community for the management and dissemination of digital materials created by the institution and its community members. It is most essentially an organizational commitment to the stewardship of these digital materials, including long-term preservation where appropriate, as well as organization and access or distribution."

Recent Trends in Libraries

Digital Transformation in Libraries

Shift from physical to digital collections.

Focus on Open Access and FAIR Principles

Findable, Accessible, Interoperable, Reusable.

Integration of Technology

AI, machine learning, and big data analytics in library operations.

Emphasis on Collaboration and Sharing

Institutional and global partnerships for open data sharing.

Developing an institutional repository

- Steps in the development of an institutional repository include **choosing a platform and defining metadata practices**.
- Setting up an IR requires **working with faculty** to identify the **type of content** the library needs to support.
- **Promoting** the IR is important to enhance access and increase the visibility of the researchers.
- **Libraries** will also need to target their marketing efforts to different groups of **stakeholders**. They may generate faculty interest by describing how an IR can support research or improve future findability of articles

Why IR

- **For the Individual**
  - Provides a central archive of their work
  - Increases the dissemination and impact of their research
  - Acts as a full CV
- **For the Institution**
  - Increases visibility and prestige
  - Acts as an advertisement to funding sources
  - Can attract new staff and students, etc.
- **For Society**
  - Provides access to the world's research (with OAI) (thus realizing Harnad's dream)
  - Ensures long-term preservation of institutes' academic output

They may coexist with subject oriented repositories!

Benefits for the Researcher

- Increase visibility of research (Papers) output
- Help in increasing citation
- Rapid Dissemination of research output
- Provide ease of access
- Cross-searchable
- Value added services
  - hit counts on papers

IR Software

Open-Source

- DSpace
- EPrints
- Fedora
- Greenstone

Why DSpace

- DSpace was developed based on the experience gained by EPrints.
- Its is open source
- It has a well defined data model:

  Community - Collection - Item - Metadata - Bundle - Bitstream
- UTF-8 capable.
- Supports various metadata schema.

Requirement For Installation Software

- Operating System (Linux)
- DSpace installation documentation:
  - http://dspace.org/technology/system-docs/install.html

Cont…

From DSpace 9 installation is divide into two parts:

- Backend
- Frontend

Note*: We can installation Backend and Frontend in different system.*

Backend Requirements

- Backend Requirements
  - UNIX-like OS or Microsoft Windows
  - Java JDK 17 (OpenJDK or Oracle JDK)
  - Apache Maven 3.8.x or above (Java build tool)
    - Configuring a Maven Proxy
  - Apache Ant 1.10.x or later (Java build tool) Apache Ant is required for the second stage of the build process (deploying/installing the application)
  - Relational Database (PostgreSQL)
    - PostgreSQL 14.x or 15.x (with pgcrypto installed)
  - Apache Solr 9.x (full-text index/search service)
  - (Optional) Servlet Engine (Apache Tomcat 10.1.x, Jetty, Caucho Resin or equivalent)
  - (Optional) IP to City Database for Location-based Statistics
  - Backend Installation

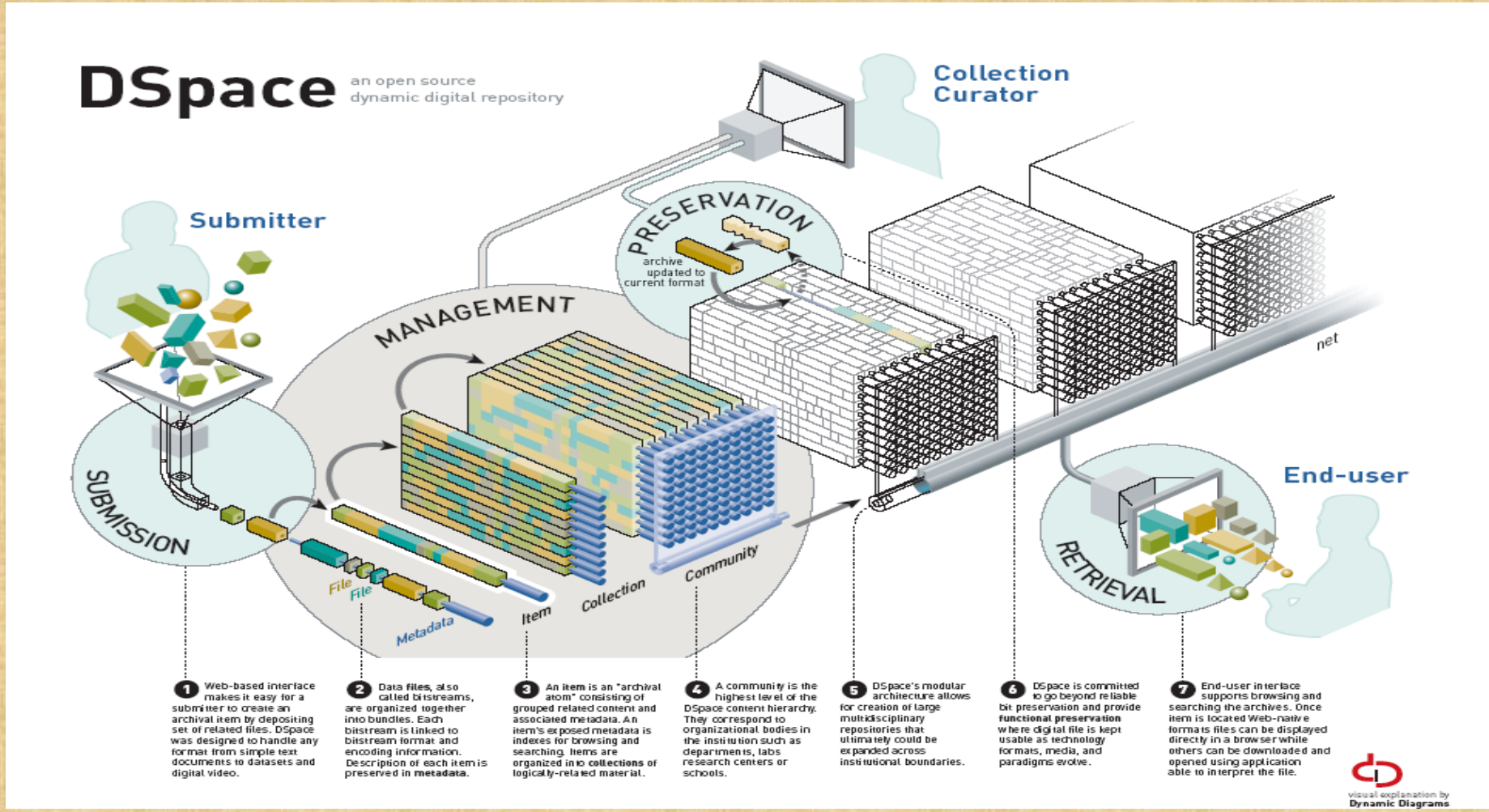
Frontend Requirements

- Installing the Frontend (User Interface)
  - UNIX-like OS or Microsoft Windows
  - Node.js (v18.x or v20.x) Node.js is a Javascript runtime that also provides npm (Node Package Manager). It is used to both build and run the frontend.
  - Yarn (v1.x)
  - PM2 (or another Process Manager for Node.js apps) (optional, but recommended for Production)
  - DSpace Backend (see above)

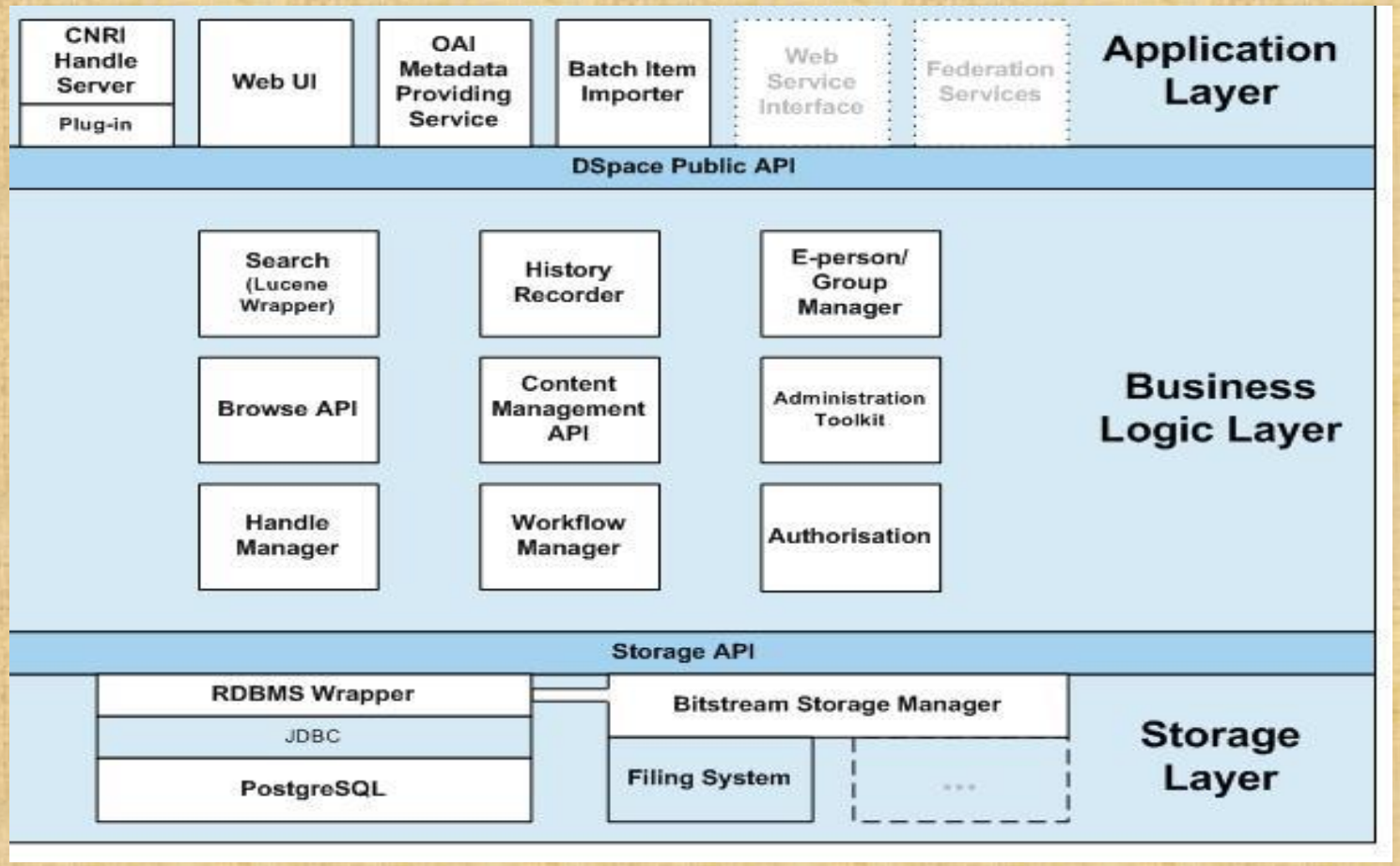
Demo

Click here for Demo

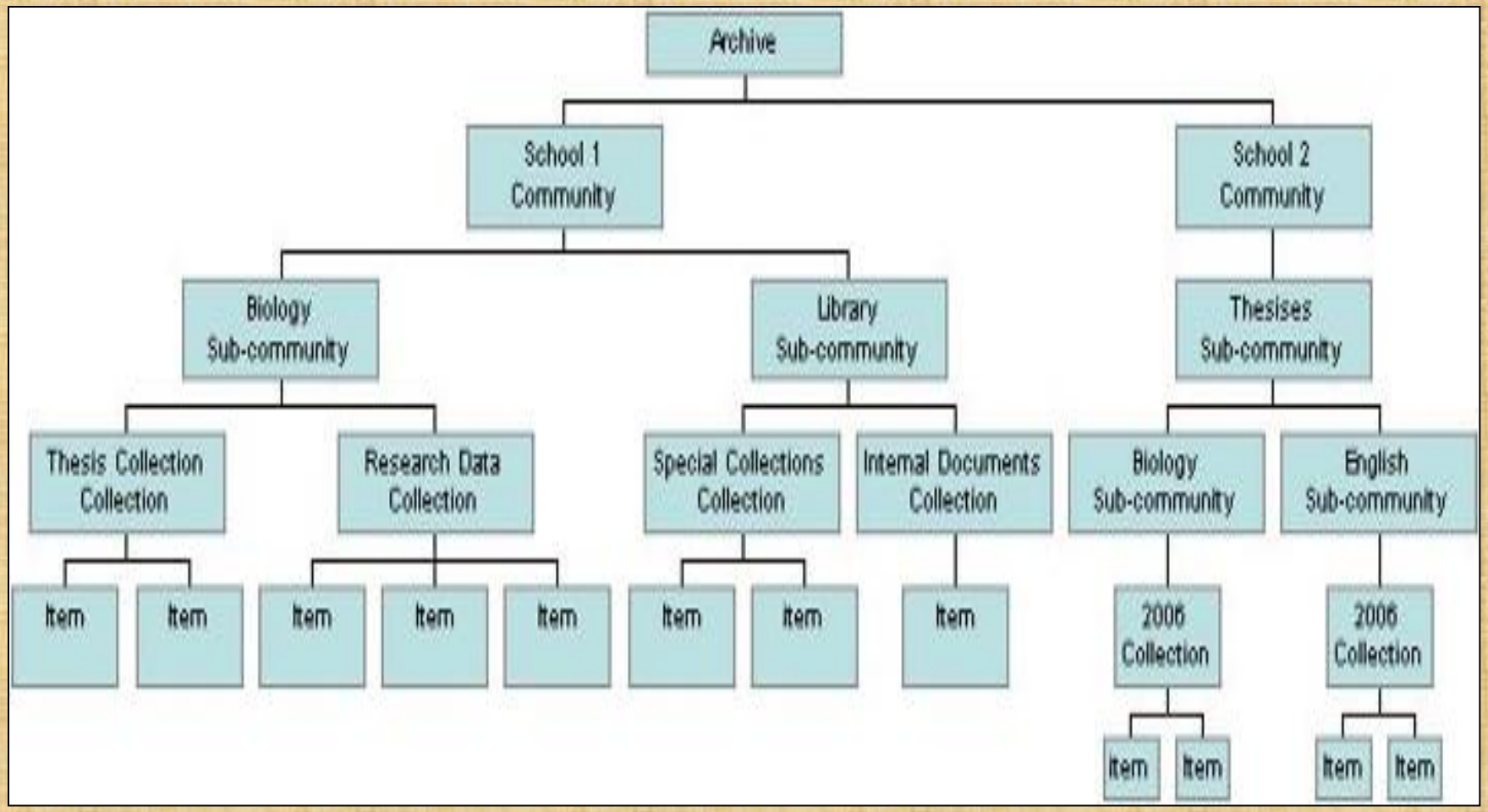
DSpace


DSpace
an open source
dynamic digital repository
Collection
Curator
Submitter
PRESERVATION
archive
updated to
current format
MANAGEMENT
net
End-user
SUBMISSION
RETRIEVAL
File
File
Metadata
Item
Collection
Community
1 Web-based interface makes it easy for a submitter to create an archival item by depositing set of related files. DSpace was designed to handle any format from simple text documents to datasets and digital video.
2 Data files, also called bitstreams, are organized together into bundles. Each bitstream is linked to bitstream format and encoding information. Description of each item is preserved in metadata.
3 An item is an "archival atom" consisting of grouped related content and associated metadata. An item's exposed metadata is indexes for browsing and searching. Items are organized into collections of logically-related material.
4 A community is the highest level of the DSpace content hierarchy. They correspond to organizational bodies in the institution such as departments, labs research centers or schools.
5 DSpace's modular architecture allows for creation of large multidisciplinary repositories that ultimately could be expanded across institutional boundaries.
6 DSpace is committed to go beyond reliable bit preservation and provide functional preservation where digital file is kept usable as technology formats, media, and paradigms evolve.
7 End-user interface supports browsing and searching the archives. Once item is located Web-native formats files can be displayed directly in a browser while others can be downloaded and opened using application able to interpret the file.
visual explanation by
Dynamic Diagrams

DSpace Architecture



Dspace sample



Communities and Collections in DSpace

- Chemistry
  - CHEM Conference Papers
  - CHEM Journal/Magazine Articles
  - CHEM Patents
  - CHEM Theses
- Civil Engineering
  - CIVL Conference Papers
  - CIVL Journal/Magazine Articles
  - CIVL Patents
  - CIVL Theses
- College of Lifelong Learning (CL3)
  - CL3 Conference Papers
- Computer Science
  - COMP Conference Papers
  - COMP Journal/Magazine Articles
  - COMP Patents
  - COMP Preprints
  - COMP Presentations
  - COMP TCSC Research Reports
  - COMP Technical Reports
  - COMP Theses
  - COMP Working Papers
- Cyberspace Center
  - CC Journal/Magazine Articles
  - CC Others
  - CC Research Reports

Community

Collections

HKUST Institutional Repository >
Biology >
BIOL Patents >

An Item in DSpace

Please use this identifier to cite or link to this item: http://hdl.handle.net/1783.1/315

Title: Fritillaria species identification

Authors: Tsim, Karl Wah-Keung
Ip, Nancy Y.
Sucher, Nikolaus J.

Keywords: Fritillaria

Issue Date: 27-May-2003

Citation: US Patent 6,569,625. Washington, DC: US Patent and Trademark Office, 2003.

Abstract: According to the present invention there is provided a method for determining the phylogeny of sample Fritillaria genetic material.

URI: http://hdl.handle.net/1783.1/315

Appears in Collections: BIOL Patents
BICH Patents

Metadata

A Bitstream

Files in This Item:

File	Size	Format	
US6569625.pdf	1167Kb	Adobe PDF	View/Open

Show full item record

Transpersonal Psychology 1965

Chapter 1 – Introduction to Transpersonal Psychology

Chapter 1

INTRODUCTION TO TRANSPERSONAL PSYCHOLOGY

Learning Objectives

1. Compare and contrast the perspectives of mainstream psychology and transpersonal psychology toward human spirituality.
2. List the goals of transpersonal psychology.
3. Identify key themes in definitions of transpersonal psychology cited in the literature.
4. Identify at least five definitions of "transcendence beyond ego" proposed by Abraham Maslow.
5. Describe the two types of "transcendent" abilities identified by John Curtis Gowan.
6. Define transpersonal experiences.
7. Define transpersonal behaviors.
8. Name topics studied in transpersonal psychology
9. Give examples of exceptional human experiences.
10. Tell why exceptional experiences are considered "anomalous" by mainstream psychology.
11. Describe how exceptional human experiences are viewed by transpersonal psychology.
12. Tell why experiential exercises and practices are important in learning about transpersonal concepts.
13. Distinguish between the personalistic and naturalistic approaches to the history of transpersonal psychology.
14. Explain why the founder of experimental psychology can be considered a founder of transpersonal psychology.
15. Explain why William James can be considered "the father of modern transpersonal psychology and psychiatry."
16. Describe the contribution of F. W. H. Myers's concept of the "subliminal self" to the development of transpersonal psychology.
17. Describe the role that Sigmund Freud played in the history of transpersonal psychology.
18. Describe the contribution of Alfred Adler's concept of the "creative self" to psychology.
19. List the significant contributions that C. G. Jung made in the history of transpersonal psychology.
20. Describe Roberto Assagioli's contribution to transpersonal psychology.
21. Tell how transpersonal psychology is a reflection of America's "alternative reality" and "folk psychology" traditions.
22. Describe the contribution that psychical research plays in the history of transpersonal psychology.
23. Describe the contribution of Eastern and Asian systems of thought to the development of transpersonal psychology.
24. Describe the role that the Counterculture movement played in the history of transpersonal psychology.

Importance of Open Data Repositories

- **Accessibility:** Democratizes access to data for researchers and the public.
- **Transparency:** Promotes open science and accountability.
- **Preservation:** Ensures long-term availability of valuable data.
- **Collaboration:** Facilitates interdisciplinary and cross-institutional research

Key Features of Open Data Repositories

- **Metadata Standards:** Dublin Core, DataCite for discoverability.
- **Data Preservation:** Backup, versioning, and storage strategies.
- **Persistent Identifier for Data:** Ensuring persistent citation.
- **Integration with Tools:** APIs, search engines, and visualization software.

Technologies Enabling Open Data Repositories

- **Cloud Computing:** Scalable and secure storage solutions.
- **Big Data Analytics:** Insights into usage and patterns.
- **Interoperability Standards:** Ensuring data exchange between repositories.
- **AI and ML:** Automating metadata creation and data categorization.

Libraries' Role in Repositories

- **Data Literacy Programs:** Educating users on data management and repository use.
- **Curating and Preserving Data:** Ensuring data integrity and relevance.
- **Metadata Support:** Assisting researchers with standards and best practices.
- **Advocating for Open Access:** Promoting policies and partnerships.

Challenges of IR

Starting an IR involves several issues which are to be studied and decisions have to be taken. They are:

- Policy decisions
- Constant Monitoring
- Organizational issues
- Cultural Issues

Challenges and Issues

Faculty:

- Content(Where there's will there's way,(College and Research Library news 2009)
- Content-yours, theirs, and ours(West, Computer in Lib, 2008)
- Low awareness of Open Access Initiative (OAI)
- Concern over copyright issues
- Apathy in direct submission
- Lack of willingness to negotiate on non-exclusive rights and to publish in open access journals
- Lack of willingness to provide the right versions of documents (pre- or post-refereed)
- Your Participation –Today and Tomorrow.

Challenges and Opportunities

- **Challenges:**
  - Data privacy and security concerns.
  - Compliance with diverse standards.
  - Sustainable funding models.
- **Opportunities:**
  - Enhanced collaboration between libraries and research institutions.
  - New services for data management and curation.
  - Advancing the open science movement.

Universities / Institutions:

- Needs to make a mandate to deposit all research outputs with the Institutional Repository

- Needs to give financial support to faculty who submit papers to open access journals

The Future of Libraries is Repositories

- **Expanding Roles:** Libraries as hubs for data innovation.
- **Emerging Trends:** AI-driven repositories, blockchain for data security.
- **Global Collaboration:** Strengthening networks for knowledge sharing.

Use Cases and Success Stories

- **CKAN:** Used by Data.gov for open government data.
- **Dataverse:** Widely adopted by academic institutions like Harvard University.
- **Dryad:** Supports leading scientific publishers for data sharing.
- **DKAN:** Powers civic data platforms for city governments.
- **Nesstar:** Enhances access to European Social Survey data.

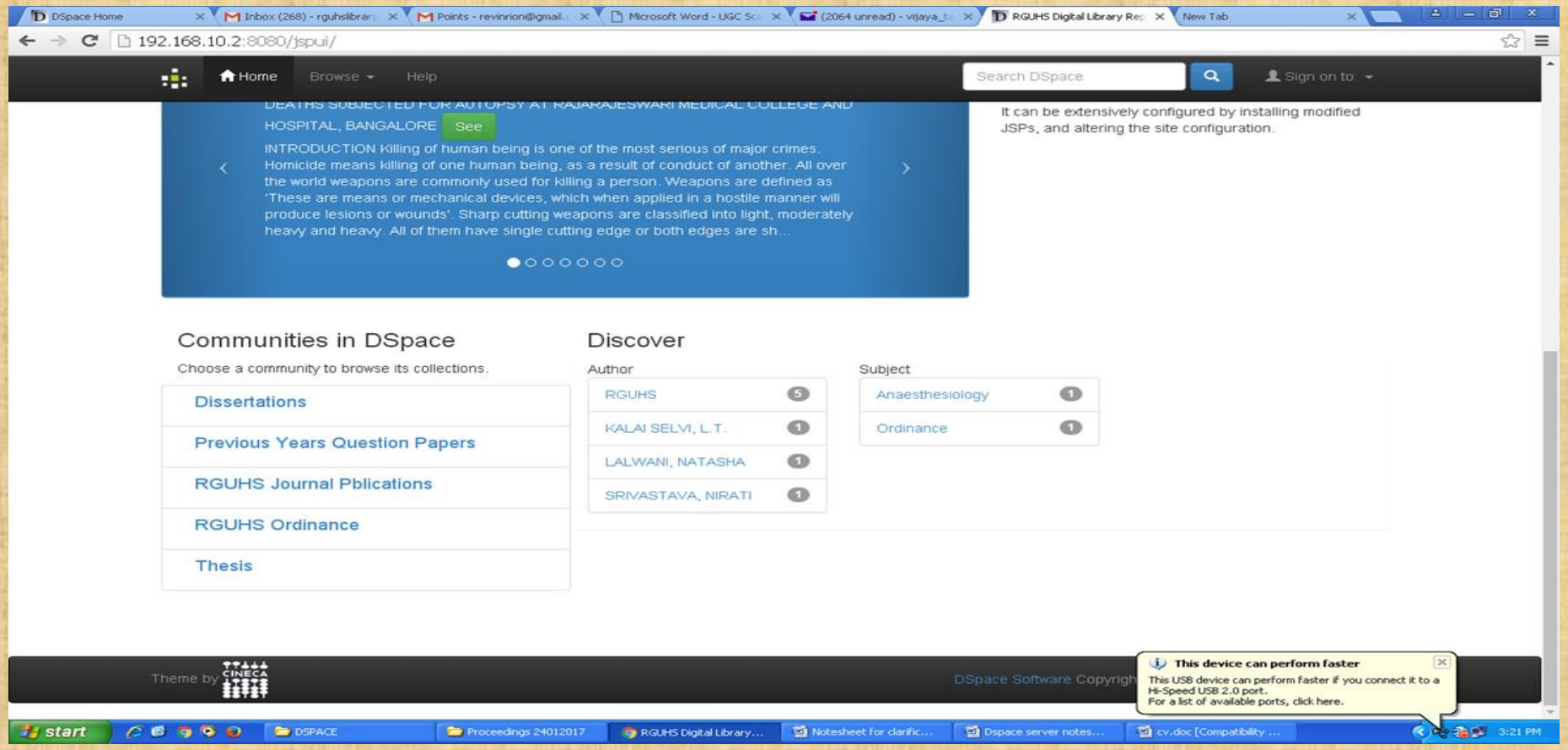
SPACE
UBC's Information Repository

Exploring Global Connections
ETD2017
Washington, DC
August 7-9, 2017
NDLTD

Success Stories (Few)

- Ch. Charan Singh University, Meerut
- Department of Library and Information Science, Bangalore University,
- ECIL, Hyderabad,
- Kuvempu University, Shimoga
- Reva Institute of Technology, Bangalore,
- National Institute of Advanced Studies, Bangalore,
- Nitte Meenakshi Institute of Technology, Bangalore,
- Sri Devarajurs Medical College, Kolar,
- R.L.Jalappa Institute of Technology, Doddaballapur,
- Rajiv Gandhi University of Health Sciences, Bangalore ,
- Ramaiah Medical College, Bangalore,
- University of Madras,

DSpace Success Stories



SDUHAE

DSpace JSPUI

DSpace preserves and enables easy and open access to all types of digital content including text, images, moving images, mpegs and data sets

Learn More

DSPACE

Digital Library at SDUAHER

Search

Search: All of DSpace

for Go

Current filters: Author Equals

2018/April/13 X

Start a new search

Add filters:

Use filters to refine the search results.

Title Equals

Add

Results/Page 10 | **Sort items by** Relevance **In order** Descending **Authors/record** All Update

Discover

Subject

Apr-13 3

RGUHS Convocation day 1

Date issued

2018 3

2019 1

Has File(s)

true 4

DELL

Dspace worskhop_n.jpg
National Workshop on KOHA, DSPACE, JOOM
Date: 18th to 20th August 2012
Aishwarya Institute of Management Studies and Research, Bangalo

NTRE
S. Altaf Ahmed

DELL

Conclusion

- Dataset-specific repository software enhances data management and sharing.
- Libraries play a critical role in supporting these technologies.
- Collaboration and innovation are essential for future success.

Thank you
for the Patience